

MATH3026/4022 Lecture 8

Model Identification and Fitting II

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- ▶ Parameter estimation for the $AR(p)$ process using the Yule-Walker equations.
- ▶ Maximum likelihood estimation for the $ARMA(p, q)$ process.
- ▶ Parameter estimation for the $MA(1)$ process:
 - ▶ Method of moments
 - ▶ Maximum likelihood estimation
 - ▶ Transformation method

During Lecture 7 we introduced parameter estimation for time series models, using method of moments and maximum likelihood estimation.

In particular, we demonstrated these methods using an $AR(1)$ model as an example.

In this lecture, we shall extend these methods and explore estimation for more general models.

$AR(p)$ process: Parameter Estimation

To begin, we briefly cover parameter estimation for the $AR(p)$ process.

Recall that the general $AR(p)$ process is

$$(X_t - \mu) = \phi_1(X_{t-1} - \mu) + \phi_2(X_{t-2} - \mu) + \dots + \phi_p(X_{t-p} - \mu) + Z_t.$$

where Z_t is a mean zero white noise process with variance σ_z^2 .

Here, we would like to estimate $p + 2$ parameters $\mu, \phi_1, \dots, \phi_p, \sigma_z^2$.

Unfortunately, it is much more difficult to obtain equations for the MLEs than was the case with the $AR(1)$ model.

$AR(p)$ Process: Likelihood Function

Suppose we assume that the white noise process $\{Z_t : t \in \mathbb{Z}\}$ is a **Gaussian process** with mean zero (i.e. all finite dimensional distributions of $\{Z_t\}$ are normal).

Then we can deduce that $(X_1, \dots, X_n)^\top$ – a sample of size n from the time series $\{X_t\}$ – has an n -dimensional multivariate normal distribution and construct a likelihood function.

To find the maximum likelihood estimates of model parameters, we would obtain a set of $p + 2$ score equations and solve these.

An analytical solution to these equations is not straightforward.

However, the `arima` command in R may be used to obtain estimates using a numerical method.

Model Fitting using R

For example, suppose that x is a set of observed time series data to which we'd like to fit a model.

Let's assume that we'd like to fit an $AR(2)$ model to these data.

Then, in R, we'd run the command:

```
ar-model<-arima(x,order=c(2,0,0),method="ML")
```

With output

Coefficients:

	ar1	ar2	intercept
	0.0709	0.2381	0.0069
s.e.	0.0307	0.0307	0.0462

σ^2 estimated as 1.021: log likelihood = -1429.54,
aic = 2867.09

$AR(p)$ process: Parameter Estimation

We can also use the Yule-Walker equations to obtain parameter estimates.

Recall the Yule-Walker equations (below, $\rho_h = \rho(h)$)

$$\rho_1 = \phi_1 + \rho_1\phi_2 + \dots + \rho_{p-1}\phi_p$$

$$\rho_2 = \phi_1\rho_1 + \phi_2 + \dots + \rho_{p-2}\phi_p$$

$$\vdots \quad \vdots$$

$$\rho_p = \phi_1\rho_{p-1} + \phi_2\rho_{p-2} + \dots + \phi_p.$$

$AR(p)$ process: Parameter Estimation

We can write these in matrix/vector form as

$$\boldsymbol{\rho}_p = \mathbf{P}_p \boldsymbol{\phi}_p \quad (8.1)$$

where $\boldsymbol{\rho}_p = (\rho_1, \dots, \rho_p)^\top$, $\boldsymbol{\phi}_p = (\phi_1, \dots, \phi_p)^\top$ and

$$\mathbf{P}_p = \begin{bmatrix} 1 & \rho_1 & \rho_2 & \dots & \rho_{p-1} \\ \rho_1 & 1 & & & \\ \rho_2 & & 1 & & \\ \vdots & & & \ddots & \rho_1 \\ \rho_{p-1} & & & \rho_1 & 1 \end{bmatrix}.$$

$AR(p)$ process: Parameter Estimation

We can substitute the unknown quantities in (8.1) with their corresponding sample estimates.

That is, replace $\boldsymbol{\rho}_p$ with $\mathbf{r}_p$ and P_p with R_p to obtain the equation

$$\mathbf{r}_p = R_p \hat{\boldsymbol{\phi}}_p.$$

Here $\mathbf{r}_p = (r_1, \dots, r_p)^\top$ and

$$R_p = \begin{bmatrix} 1 & r_1 & r_2 & \dots & r_{p-1} \\ r_1 & 1 & & & \\ r_2 & & 1 & & \\ \vdots & & & \ddots & r_1 \\ r_{p-1} & & & r_1 & 1 \end{bmatrix},$$

where r_h is the sample autocorrelation at lag h .

$AR(p)$ process: Parameter Estimation

Then the estimates for ϕ are

$$\hat{\phi}_p = R_p^{-1} \mathbf{r}_p.$$

μ can be estimated using the sample mean

$$\hat{\mu} = \bar{x} = \frac{1}{n} \sum_{t=1}^n x_t.$$

σ_z^2 can be estimated as

$$\hat{\sigma}_z^2 = c_0 \left[1 - \hat{\phi}_1 r_1 - \hat{\phi}_2 r_2 - \dots - \hat{\phi}_p r_p \right].$$

where

$$c_0 = \frac{1}{n} \sum_{t=1}^n (x_t - \bar{x})^2.$$

Example 8.1

Consider the (weakly) stationary AR(1) model for the process $\{X_t\}$, written

$$X_t - \mu = \phi_1 (X_{t-1} - \mu) + Z_t$$

where $\{Z_t\}$ is a white noise process with variance σ_z^2 .

Assuming that a sample of size n of $\{X_t\}$ is to be observed. Using the Yule-Walker equations, determine the estimators of μ , ϕ_1 and σ_z^2 .

Example 8.1

Example 8.2

Suppose that a sample of data is observed from the $AR(2)$ process

$$X_t = \phi_1 X_{t-1} + \phi_2 X_{t-2} + Z_t$$

Using the Yule-Walker equations determine the estimates of ϕ_1 , ϕ_2 and σ_z^2 .

Example 8.2

$ARMA(p, q)$ process: Maximum Likelihood

To consider maximum likelihood methods for an $ARMA(p, q)$ process, we first recall the multivariate normal distribution.

Suppose that $\mathbf{x} = (x_1, \dots, x_n)^\top$ is a set of observations from a multivariate normal distribution with dimension n .

That is, $\mathbf{x}$ are observed values of the random variables $\mathbf{X} = (X_1, \dots, X_n)^\top$ where

$$\mathbf{X} \sim \mathcal{N}(\boldsymbol{\mu}_n, \Gamma_n).$$

Here $\boldsymbol{\mu}_n$ is a vector of dimension n such that $\mathbb{E}(\mathbf{X}) = \boldsymbol{\mu}_n$ and Γ_n is the $n \times n$ positive definite variance-covariance matrix of $\mathbf{X}$.

$ARMA(p, q)$ process: Maximum Likelihood

Now we shall assume further that $\mathbf{X} = (X_1, \dots, X_n)^\top$ are n sequential observations from a stationary time series such that

$$\mathbb{E}(\mathbf{X}) = \boldsymbol{\mu}_n = (\mu, \dots, \mu)^\top$$

In other words, $\boldsymbol{\mu}_n$ is a vector of dimension n where each element is equal to μ .

Also, the variance-covariance matrix of $\mathbf{X}$ is Γ_n , where

$$\Gamma_n = \begin{bmatrix} \gamma_0 & \gamma_1 & \gamma_2 & \dots & \gamma_{n-1} \\ \gamma_1 & \gamma_0 & & & \\ \gamma_2 & & \gamma_0 & & \\ \vdots & & & \ddots & \\ \gamma_{n-1} & & & & \gamma_0 \end{bmatrix}.$$

$ARMA(p, q)$ process: Maximum Likelihood

Recall the $ARMA(p, q)$ process

$$(X_t - \mu) - \phi_1(X_{t-1} - \mu) - \dots - \phi_p(X_{t-p} - \mu) = Z_t + \theta_1 Z_{t-1} + \dots + \theta_q Z_{t-q}$$

where the parameters of interest are $(\sigma_z^2, \mu, \phi_1, \dots, \phi_p, \theta_1, \dots, \theta_q)$.

We note that the elements of Γ_n are functions of $\sigma_z^2, \mu, \phi_1, \dots, \phi_p, \theta_1, \dots, \theta_q$.

Suppose we assume that the white noise terms are independent $\mathcal{N}(0, \sigma_z^2)$ random variables then the variables $X_1, \dots, X_n$ (from an $ARMA(p, q)$ process) are multivariate normal.

We'll now justify this briefly.

$ARMA(p, q)$ process - Multivariate Normal Distribution

ARMA(p, q) process: Maximum Likelihood

The probability density function of $\mathbf{X}$ is

$$f(\mathbf{x}) = (2\pi)^{-\frac{n}{2}} |\Gamma_n|^{-\frac{1}{2}} \exp \left\{ -\frac{1}{2} (\mathbf{x} - \boldsymbol{\mu})^\top \Gamma_n^{-1} (\mathbf{x} - \boldsymbol{\mu}) \right\}.$$

Moreover, denoting $\boldsymbol{\beta} = (\phi_1, \dots, \phi_p, \theta_1, \dots, \theta_q)^\top$, the likelihood function is

$$\mathcal{L}(\mu, \boldsymbol{\beta}, \sigma_z^2 | \mathbf{x}) = (2\pi)^{-\frac{n}{2}} |\Gamma_n|^{-\frac{1}{2}} \exp \left\{ -\frac{1}{2} (\mathbf{x} - \boldsymbol{\mu})^\top \Gamma_n^{-1} (\mathbf{x} - \boldsymbol{\mu}) \right\}.$$

$ARMA(p, q)$ process: Maximum Likelihood

Then the log-likelihood function is

$$\ell(\mu, \beta, \sigma_z^2 | \mathbf{x}) = -\frac{n}{2} \log(2\pi) - \frac{1}{2} \log |\Gamma_n| - \frac{1}{2} (\mathbf{x} - \mu)^\top \Gamma_n^{-1} (\mathbf{x} - \mu)$$

Unfortunately, differentiating the log-likelihood with respect to the parameters of interest and solving the resulting set of score equations to obtain the MLEs is not straightforward.

Instead, numerical optimisation is typically used to obtain maximum likelihood estimates for parameters, given a set of observed data $\mathbf{x} = (x_1, \dots, x_n)^\top$ from an $ARMA(p, q)$ process.

Parameter Estimation: MA Processes

Now we turn our attention to estimating parameters for moving average (MA) models.

We begin by considering parameter estimation for the familiar $MA(1)$ process.

We recall that the $MA(1)$ process is written

$$X_t - \mu = Z_t + \theta Z_{t-1}.$$

where $Z_t, Z_{t-1}, \dots$ are independent white noise terms with mean zero and variance σ_z^2 .

We shall assume that the process is invertible (i.e. $|\theta| < 1$).

The task here is to estimate the parameters μ, θ and σ_z^2 , having observed a set of data $\mathbf{x} = (x_1, \dots, x_n)^\top$ from an $MA(1)$ process.

Method of Moments Estimation

A simple approach to parameter estimation is to use the method of moments.

For an $MA(1)$ process, we recall that the correlation at lag 1 is

$$\rho(1) = \frac{\theta}{1 + \theta^2}.$$

Using the method of moments, we can replace $\rho(1)$ with the sample equivalent r_1 and solve for θ . That is

$$\begin{aligned} r_1 &= \frac{\tilde{\theta}}{1 + \tilde{\theta}^2} \\ \implies r_1 - \tilde{\theta} + r_1 \tilde{\theta}^2 &= 0 \end{aligned} \tag{8.2}$$

Method of Moments Estimation

Solving (8.2), we obtain

$$\tilde{\theta} = \frac{1 \pm \sqrt{1 - 4r_1^2}}{2r_1}.$$

We note that we need to impose the constraint $|\theta| < 1$ to ensure invertibility (and $|r_1| < 1/2$ for real roots).

Naturally, μ and σ_z^2 may be estimated in a similar manner. Where

$$\tilde{\mu} = \bar{x} = \frac{1}{n} \sum_{t=1}^n x_t.$$

and

$$c_0 = \tilde{\sigma}_z^2(1 + \tilde{\theta}^2) \implies \tilde{\sigma}_z^2 = \frac{c_0}{1 + \tilde{\theta}^2}.$$

Example 8.3

Suppose that $\{X_t\}$ is the $MA(1)$ process

$$X_t = \theta Z_{t-1} + Z_t$$

where $\{Z_t\}$ is a white noise process with variance σ_z^2 .

A sample of n values from this process are observed where $r_1 = -0.25$ and $c_0 = 1.25$. Determine the method of moments estimates of θ and σ_z^2 .

Example 8.3

Method of Moments Estimation

A problem with method of moments estimation here is that the method of moments estimator for θ is not a 'good' estimator.

Indeed, $\tilde{\theta}$ is typically very imprecise (i.e. has a high variance) when compare to the MLE and so we tend to avoid this approach to estimation for MA models.

Next, we consider maximum likelihood estimation for the $MA(1)$ process.

Maximum Likelihood Estimation

As usual, we require a distributional assumption in order to do maximum likelihood estimation.

Here, we shall assume that the white noise terms are independent normal random variables, each with mean zero and variance σ_z^2 .

It follows that the random variables $\mathbf{X}_n = (X_1, \dots, X_n)^\top$, from an $MA(1)$ process, have a multivariate normal distribution with mean $\boldsymbol{\mu}_n = (\mu, \dots, \mu)^\top$ (vector of dimension n with each entry μ) and variance-covariance matrix Γ_n , where

$$\Gamma_n = \begin{bmatrix} \gamma_0 & \gamma_1 & \gamma_2 & \cdots & \gamma_{n-1} \\ \gamma_1 & \gamma_0 & & & \\ \gamma_2 & & \gamma_0 & & \\ \vdots & & & \ddots & \\ \gamma_{n-1} & & & & \gamma_0 \end{bmatrix}.$$

Maximum Likelihood Estimation

For an $MA(1)$ process, we know that

$$\gamma_0 = \sigma_z^2(1 + \theta^2); \quad \gamma_1 = \theta\sigma_z^2; \quad \gamma_h = 0 \quad \text{where } |h| > 1.$$

Then

$$\Gamma_n = \sigma_z^2 \begin{bmatrix} 1 + \theta^2 & \theta & 0 & \dots & 0 \\ \theta & 1 + \theta^2 & \theta & & \\ 0 & \theta & 1 + \theta^2 & & \\ \vdots & & \ddots & \ddots & \\ \vdots & & \ddots & \ddots & \theta \\ 0 & & & \theta & 1 + \theta^2 \end{bmatrix}.$$

Maximum Likelihood Estimation

The likelihood function is

$$\mathcal{L}(\mu, \theta, \sigma_z^2 | \mathbf{x}) = (2\pi)^{-\frac{n}{2}} |\Gamma_n|^{-\frac{1}{2}} \exp \left\{ -\frac{1}{2} (\mathbf{x} - \boldsymbol{\mu})^\top \Gamma_n^{-1} (\mathbf{x} - \boldsymbol{\mu}) \right\}.$$

The log-likelihood function is

$$\ell(\mu, \theta, \sigma_z^2 | \mathbf{x}) = -\frac{n}{2} \log(2\pi) - \frac{1}{2} \log |\Gamma_n| - \frac{1}{2} (\mathbf{x} - \boldsymbol{\mu})^\top \Gamma_n^{-1} (\mathbf{x} - \boldsymbol{\mu}).$$

Once again, differentiation of the log-likelihood function and solution of the score equations is very difficult.

As such, numerical optimisation is used to obtain maximum likelihood estimates.

Transformation Method

A third method for estimation is to rewrite the $MA(1)$ model as

$$Z_t = X_t - \mu - \theta Z_{t-1}.$$

For simplicity, we define $X'_t = X_t - \mu$ then, iterating from Z_0 to Z_n , we write

$$Z_0 = Z_0$$

$$Z_1 = X'_1 - \theta Z_0$$

$$Z_2 = X'_2 - \theta Z_1$$

$$= X'_2 - \theta X'_1 + (-\theta)^2 Z_0$$

$$Z_3 = X'_3 - \theta Z_2$$

$$= X'_3 + (-\theta)X'_2 + (-\theta)^2 X'_1 + (-\theta)^3 Z_0$$

$$\vdots$$

$$\begin{aligned} Z_n &= X'_n + (-\theta)X'_{n-1} + (-\theta)^2 X'_{n-2} \\ &\quad + \dots + (-\theta)^{n-1} X'_1 + (-\theta)^n Z_0. \end{aligned}$$

Transformation Method

We see that we have a transformation (mapping) where

$(Z_0, X'_1, \dots, X'_n)$ have been transformed to $(Z_0, \dots, Z_n)$.

The Jacobian matrix for the transformation is

$$\mathcal{J} = \begin{bmatrix} \frac{\partial Z_0}{\partial Z_0} & \frac{\partial Z_0}{\partial X'_1} & \frac{\partial Z_0}{\partial X'_2} & \cdots & \frac{\partial Z_0}{\partial X'_n} \\ \frac{\partial Z_1}{\partial Z_0} & \frac{\partial Z_1}{\partial X'_1} & \frac{\partial Z_1}{\partial X'_2} & \cdots & \frac{\partial Z_1}{\partial X'_n} \\ \vdots & \vdots & \ddots & \ddots & \vdots \\ \frac{\partial Z_n}{\partial Z_0} & \frac{\partial Z_n}{\partial X'_1} & \frac{\partial Z_n}{\partial X'_2} & \cdots & \frac{\partial Z_n}{\partial X'_n} \end{bmatrix}.$$

Evaluating the Jacobian Matrix

Transformation Method

It follows that

$$\mathcal{J} = \begin{bmatrix} 1 & 0 & 0 & \dots & \dots & 0 \\ -\theta & 1 & 0 & \dots & \dots & 0 \\ (-\theta)^2 & -\theta & 1 & 0 & \dots & 0 \\ \vdots & \vdots & \ddots & \ddots & \ddots & 0 \\ (-\theta)^n & (-\theta)^{n-1} & & & -\theta & 1 \end{bmatrix}.$$

Clearly $\mathcal{J}$ is lower triangular and hence the Jacobian determinant is $|\mathcal{J}| = 1$.

The probability density function of the transformed distribution $(Z_0, X'_1, \dots, X'_n)^\top$ is

$$f_{Z_0, \mathbf{X}'}(z_0, \mathbf{x}') = |\mathcal{J}| f_{\mathbf{Z}}(\mathbf{z}).$$

Where $\mathbf{Z} = (Z_0, Z_1, \dots, Z_n)^\top$ with $\mathbf{z} = (z_0, z_1, \dots, z_n)^\top$ and $\mathbf{X}' = (X'_1, \dots, X'_n)^\top$ with $\mathbf{x}' = (x'_1, \dots, x'_n)^\top$.

Transformation Method

Since the white noise terms are independent, we note that the joint density of $\mathbf{Z}$, written $f_{\mathbf{Z}}(\mathbf{z})$, is multivariate normal with mean $\mathbf{0}$ and variance-covariance matrix $\sigma_z^2 \mathbf{I}$ (here $\mathbf{I}$ is the $(n+1) \times (n+1)$ identity matrix).

Then we obtain

$$\begin{aligned} f_{Z_0, \mathbf{X}'}(z_0, \mathbf{x}') &= (2\pi\sigma_z^2)^{-\frac{n}{2}} \exp \left\{ -\frac{1}{2\sigma_z^2} \mathbf{z}^\top \mathbf{z} \right\} \\ &= (2\pi\sigma_z^2)^{-\frac{n}{2}} \exp \left\{ -\frac{1}{2\sigma_z^2} \sum_{t=0}^n z_t^2 \right\}. \end{aligned}$$

Here, we note that $\mathbf{z}$ is a transformation of $(z_0, \mathbf{x}')$.

We see that it is easier to work with the joint density of $\mathbf{Z}$ since $Z_0, \dots, Z_n$ are independent.

Transformation Method

Unfortunately, whilst we may have collected observations $x_1, \dots, x_n$, we would not have a sampled value of z_0 .

We might take two possible approaches here

1. Replace z_0 by $\mathbb{E}(Z_0) = 0$, then replace $\sum_{t=0}^n z_t^2$ with $\sum_{t=1}^n z_t^2$ in the likelihood function and maximise.
2. Treat z_0 as an additional unknown parameter and then find z_0, θ, μ and σ_z^2 that maximise the likelihood function.

Although the likelihood function appears straightforward, we note that $\sum_{t=0}^n z_t^2$ is a non-linear function of θ .

As such, we would use numerical methods to obtain the maximum likelihood estimates.

Revisiting Estimation: $ARMA(p, q)$ models

In general, the most common approach used for parameter estimation in time series models is maximum likelihood estimation, typically using numerical optimisation methods.

In R estimation will typically involve using the `arima` command.

We'll cover how time series models are fitted during the MATH3026/MATH4022 R computer practicals.

Lecture 8: Learning Outcomes

- ▶ Understand how parameters for the $AR(p)$ process may be estimated using the Yule-Walker equations.
- ▶ Understand how the likelihood function may be constructed for the $ARMA(p, q)$ model and used to estimate model parameters.
- ▶ Understand how the method of moments, maximum likelihood and transformation approaches may be used to estimate parameters for the $MA(1)$ process.